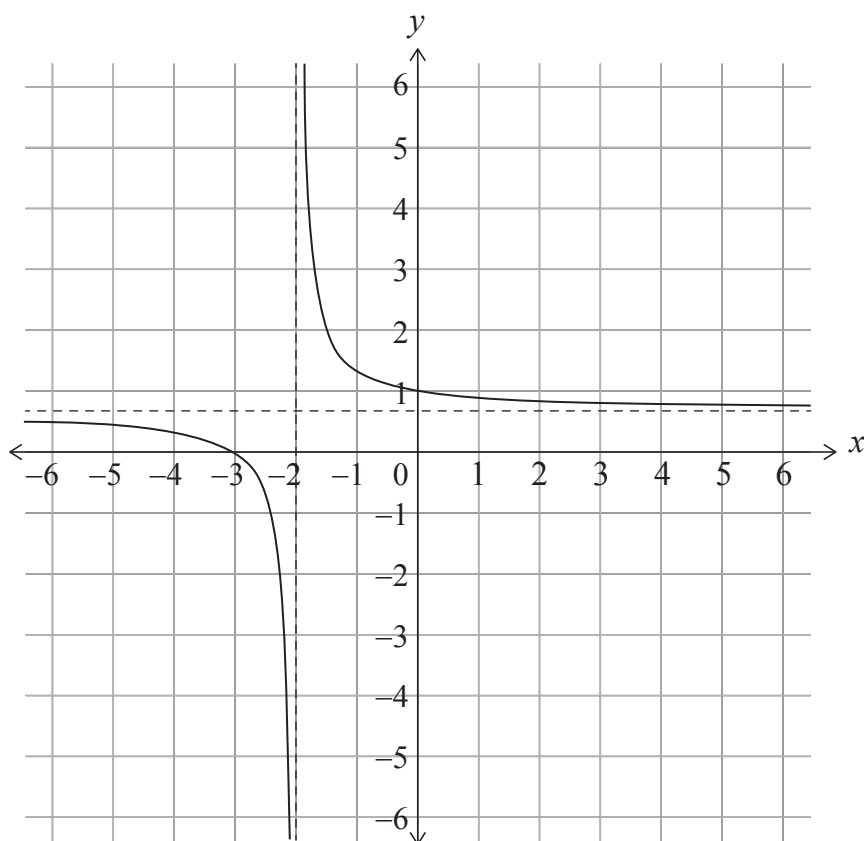


5. [Maximum mark: 8]

A function  $f$  is defined by  $f(x) = \frac{2(x+3)}{3(x+2)}$ , where  $x \in \mathbb{R}, x \neq -2$ .

The graph  $y = f(x)$  is shown below.



(a) Write down the equation of the horizontal asymptote.

[1]

Consider  $g(x) = mx + 1$ , where  $m \in \mathbb{R}, m \neq 0$ .

(b) (i) Write down the number of solutions to  $f(x) = g(x)$  for  $m > 0$ .

(ii) Determine the value of  $m$  such that  $f(x) = g(x)$  has only one solution for  $x$ .

(iii) Determine the range of values for  $m$ , where  $f(x) = g(x)$  has two solutions for  $x \geq 0$ .

[7]

(This question continues on the following page)



(Question 5 continued)

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Turn over